WORKED SOLUTION

STEP III 2009, Question 8

Pure • Limits and Reduction Formulae

STEP III 2009, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q8

This question has a full guided treatment in *Beyond STEP*, Section 4.5, *The Sophomore's Dream*: https://beyondstep.keithmcnulty.org/ch04/sophomores-dream.html

The key idea

Every part is about the logarithm being "weak": near 0, $\ln x$ blows up so slowly that any positive power of x beats any power of $\ln x$. The substitution $x = \mathrm{e}^{-t}$ turns this into the given fact about exponentials beating powers. That limit then kills the boundary terms when we integrate by parts. In the last part, the series for the exponential, applied to $x^x = \mathrm{e}^{x \ln x}$, produces exactly the integrals of part (ii).

Solution

Part (i)

Since $\ln x$ is defined only for $x > 0$, the limit means $x \to 0^+$. Put $x = \mathrm{e}^{-t}$, so that $\ln x = -t$, and $x \to 0^+$ exactly when $t \to \infty$. Then

$$x^m (\ln x)^n = \mathrm{e}^{-mt}(-t)^n = (-1)^n \mathrm{e}^{-mt} t^n.$$

The factor $(-1)^n$ is just a fixed sign, and $\mathrm{e}^{-mt} t^n \to 0$ as $t \to \infty$ by the given result. Hence

$$\lim_{x \to 0} x^m (\ln x)^n = 0.$$

For the second result, take $m = n = 1$: this gives $x \ln x \to 0$ as $x \to 0$. Since the exponential function is continuous,

$$x^x = \mathrm{e}^{x \ln x} \to \mathrm{e}^0 = 1 \qquad \text{as } x \to 0,$$

as required.

Part (ii)

First, the integral makes sense. The integrand $x^m(\ln x)^n$ is continuous on $(0,1]$, and by part (i) it tends to 0 as $x \to 0$. So it is bounded on $(0,1]$, and I_n is an ordinary (convergent) integral.

Integrate I_{n+1} by parts with $u = (\ln x)^{n+1}$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = x^m$, so that $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{(n+1)(\ln x)^n}{x}$ and $v = \dfrac{x^{m+1}}{m+1}$:

$$\begin{aligned} I_{n+1} &= \left[\frac{x^{m+1}(\ln x)^{n+1}}{m+1}\right]_0^1 - \int_0^1 \frac{x^{m+1}}{m+1} \cdot \frac{(n+1)(\ln x)^n}{x}\,\mathrm{d}x \\ &= \left[\frac{x^{m+1}(\ln x)^{n+1}}{m+1}\right]_0^1 - \frac{n+1}{m+1}\int_0^1 x^m(\ln x)^n\,\mathrm{d}x. \end{aligned}$$

? **Ask yourself** What is the boundary term at $x = 0$, where $\ln x$ isn't even defined?

The lower limit really means the limit as $x \to 0^+$. By part (i), with $m+1$ and $n+1$ in place of m and n (both allowed, since $m+1 \geq 1$ and $n+1 \geq 0$), $x^{m+1}(\ln x)^{n+1} \to 0$. At the upper limit, $\ln 1 = 0$ and $n+1 \geq 1$, so that term is 0 too. The bracket therefore vanishes, and

$$I_{n+1} = -\frac{n+1}{m+1}\,I_n,$$

as required.

To evaluate I_n, start from

$$I_0 = \int_0^1 x^m\,\mathrm{d}x = \frac{1}{m+1}.$$

Each step of the recurrence multiplies by $-\dfrac{k}{m+1}$ for $k = 1, 2, \ldots, n$, so

$$I_n = \left(-\frac{1}{m+1}\right)\left(-\frac{2}{m+1}\right)\cdots\left(-\frac{n}{m+1}\right)\frac{1}{m+1} = \frac{(-1)^n\,n!}{(m+1)^{n+1}}.$$

(A one-line induction confirms this: if it holds for n, the recurrence gives $-\frac{n+1}{m+1} \cdot \frac{(-1)^n n!}{(m+1)^{n+1}} = \frac{(-1)^{n+1}(n+1)!}{(m+1)^{n+2}}$.)

Part (iii)

For $0 < x \leq 1$, write $x^x = \mathrm{e}^{x\ln x}$ and use the exponential series:

$$x^x = \sum_{n=0}^{\infty} \frac{(x\ln x)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n(\ln x)^n}{n!}.$$

(The value at $x = 0$ does not affect the integral; by part (i) the function x^x tends to 1 there, so it is bounded and the integral is an ordinary one.)

Now integrate term by term. The $n = 0$ term is $\int_0^1 1\,\mathrm{d}x = 1$. For $n \geq 1$, the nth term is I_n from part (ii) with $m = n$ (allowed, since then m is a positive integer):

$$\frac{1}{n!}\int_0^1 x^n (\ln x)^n\,\mathrm{d}x = \frac{1}{n!}\cdot\frac{(-1)^n\,n!}{(n+1)^{n+1}} = \frac{(-1)^n}{(n+1)^{n+1}}.$$

This formula gives 1 when $n = 0$ as well. So

$$\int_0^1 x^x\,\mathrm{d}x = \sum_{n=0}^{\infty}\frac{(-1)^n}{(n+1)^{n+1}} = 1 - \left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{3}\right)^3 - \left(\tfrac{1}{4}\right)^4 + \cdots,$$

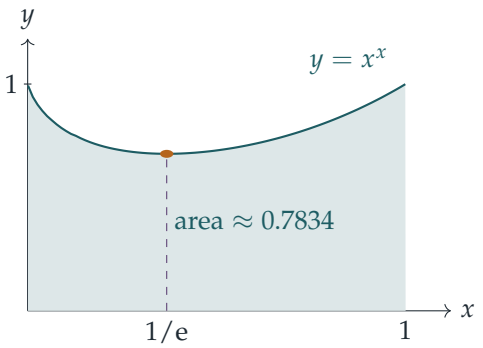
which is the required result.

Ask yourself Are we allowed to integrate an infinite sum term by term?

Not automatically, but here it is safe because $x \ln x$ stays small. Its derivative is $\ln x + 1$, so on $(0, 1]$ it has its minimum at $x = 1/\mathrm{e}$, and $|x \ln x| \leq 1/\mathrm{e} < 1$. If we stop the series after the u^N term, with $u = x \ln x$, the remainder is

$$\left|\sum_{n>N}\frac{u^n}{n!}\right| \leq \sum_{n>N}\frac{|u|^n}{n!} \leq \sum_{n>N}\frac{1}{n!} \leq \frac{2}{(N+1)!}$$

for every x in $(0, 1]$. Integrating over an interval of length 1, the error in the truncated term-by-term integral is at most $2/(N+1)!$, which tends to 0. So the integral equals the sum of the series.



Checks

- Take $m = 1$, $n = 1$ in part (ii): $\int_0^1 x \ln x\,\mathrm{d}x = -\frac{1}{4}$, which matches direct integration by parts ($[\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2]_0^1 = -\frac{1}{4}$).
- Numerically, $\int_0^1 x^x\,\mathrm{d}x \approx 0.7834305$ by the midpoint rule, and the partial sums $1 - \frac{1}{4} + \frac{1}{27} - \frac{1}{256} + \frac{1}{3125} - \cdots$ settle to 0.7834305 within a handful of terms, since the terms shrink extremely fast.
- The graph sits between its minimum $\mathrm{e}^{-1/\mathrm{e}} \approx 0.69$ and 1, so the area must be between 0.69 and 1, as it is.

Common slips

- Dropping the boundary term at 0 without comment. It is the whole point of part (i), and it needs $m + 1 \geq 1$.
- Getting the sign wrong in I_n: $\ln x < 0$ on $(0, 1)$, so I_n has the sign of $(-1)^n$.
- Forgetting the $n = 0$ term in part (iii), which is where the leading 1 comes from.

Beyond STEP

The final identity is known as the "sophomore's dream", found by Johann Bernoulli in 1697. The step we had to justify, swapping an integral and an infinite sum, is a central theme of first-year analysis: it is safe when the series converges *uniformly*, which is what our remainder bound shows (it is an instance of the Weierstrass M-test). Abbott's *Understanding Analysis* gives a very readable account of uniform convergence and when term-by-term integration works.

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